NOTATION · CHEAT SHEET

Combinatorics Notation at a Glance

The marks from the notation cards across the combinatorics pages: how each is read and what it means. Every group links to the full card on its page.

PERMUTATIONS

$n!$	**N FACTORIAL** $n \times (n-1) \times \cdots \times 1$: the count of complete rearrangements.
$P(n,r) \cdot {}_nP_r$	**P N R; N PERMUTE R** Ordered selections of r from n: $n!/(n-r)!$.
$!n \cdot D_n$	**SUBFACTORIAL N; DERANGEMENTS OF N** Permutations that leave nothing in its place.

BINOMIAL COEFFICIENT

$\binom{n}{k}$	**N CHOOSE K** Unordered selections of k from n; no bar between the numbers.
$C(n,k) \cdot {}_nC_k$	**C N K; N CHOOSE K ON A CALCULATOR** The linear spellings: introductory texts, and calculator keys.
C_n^k vs C_k^n	**C WITH N AND K IN THE CORNERS** The European forms; which corner holds n varies by tradition.

BINOMIAL THEOREM

$\sum_{k=0}^{n}$	**THE SUM AS K RUNS FROM ZERO TO N** Sigma folds the $n+1$ terms of the expansion into one expression.
T_{k+1}	**THE K-PLUS-FIRST TERM** The term containing b^k; numbering starts at T_1.
$\sum_{k_1+\cdots+k_r=n}$	**THE SUM OVER ALL SPLITTINGS OF N** One term for every way of writing n as r non-negative parts.

SAME MARKS, DIFFERENT JOB

$\binom{n}{k}$	A count; not a fraction with its bar missing, and not a column vector.
$P(n,r)$ vs $P(A)$	A permutation count here; a probability in probability.
$!n$ vs $n!$	The position decides: derangements before the number, factorial after it.
C_n^k	Read the corners by the book's convention; n may sit on top or below.